4.4 Trigonometric Form of a Complex Number

Name _____ Period ____ Date _____

Pre-Calculus 4.4 (30-minute plan) · Student sheet

I can…

- write a complex number in trigonometric form.
- write a complex number in trigonometric form in standard form.
- multiply and divide complex numbers in trigonometric form.

Vocabulary · fill in the missing word

Trigonometric form — z = _____

Modulus — r = _____ = $\sqrt{(a^2 + b^2)}$

Argument — the _____ θ, where $\tan\theta = \frac{b}{a}$

Part 1 · Notes and examples (fill in each note, then work the example)

Trigonometric form: a + bi = r(cos θ + i sin θ), with r = _____ and tan θ = _____.

Example 1

Write $-2 - 2\sqrt{3}\,i$ in trigonometric form.

Choosing θ: use the _____ of the point a + bi.

Example 2

Write $2\sqrt{2}\left(\cos\left(\frac{-\pi}{3}\right) + i\sin\left(\frac{-\pi}{3}\right)\right)$ in standard form.

Product: $z_1z_2 = r_1r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$: multiply the moduli and _____ the arguments.

Quotient: divide the moduli and _____ the arguments.

Example 3

Multiply: $2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) \cdot 8(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6})$

Part 2 · Partner practice

Partner practice

Multiply: $3(\cos 40° + i \sin 40°) \cdot 2(\cos 50° + i \sin 50°)$. Write the answer in standard form.